plantDescriptionManagement

Service Description

Service ID: *"plantDescriptionManagement"*

Abstract
This document describes the `plantDescriptionManagement` service, version 1.0.0, produced by the PlantDescriptionEngine support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `plantDescriptionManagement` service, which provides management of plant descriptions: the engineered systems of the plant, the services they consume and produce (ports), and the connections between ports. It is produced by the PlantDescriptionEngine system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 PlantDescriptionEngine management service.

2.2 How This Service Is Meant to Be Used

Engineers and LocalCloudManager create, replace, update, activate and remove plant descriptions. One description (with its includes) is active; its connections become a managed provider snapshot in Dynamic ServiceOrchestration.

2.3 Important Delimitations and Dependencies

Descriptions are validated (includes, unique systems, ports and connections, naming). Requires the local DSO serviceOrchestrationPlantManagement extension. PDE atomically replaces its complete snapshot; unresolved producers retain empty scopes, and DSO applies its normal security checks. Access follows the management policy (standalone deployment: `sysop-only`; LocalCloudManager needs the whitelist policy with `management.whitelist=LocalCloudManager`).

3 Service Interface

This section describes the interfaces to the `plantDescriptionManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

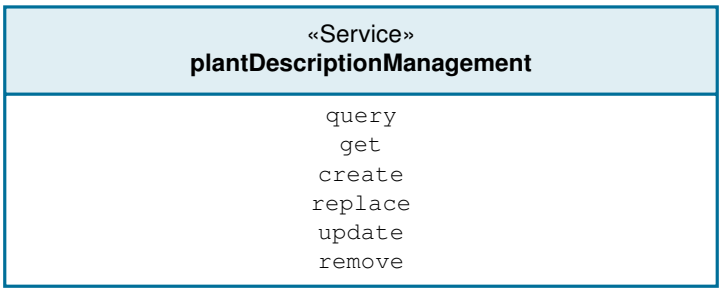


Figure 1: SysML-style block of the `plantDescriptionManagement` service and its operations.

The following interface operations are available.

3.1 operation query () : PlantDescriptionListResponse

Returns a page of plant descriptions, optionally only the active one.

3.2 operation get (Identifier) : PlantDescriptionResponse

Returns a plant description.

3.3 operation create (PlantDescriptionRequest) : PlantDescriptionResponse

Creates a plant description; if active, it replaces the active one and the rules are synchronized.

3.4 operation replace (PlantDescriptionRequest) : PlantDescriptionResponse

Replaces a plant description.

3.5 operation update (PlantDescriptionUpdate) : PlantDescriptionResponse

Changes the given fields of a plant description.

3.6 operation remove (Identifier)

Removes a plant description and its rules.

4 Information Model

Here, all data objects that can be part of the `plantDescriptionManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.9, which are used to represent things like hashes and identifiers.

4.1 struct Connection

A connection from a consumer port to a producer port.

Field	Type	Description
consumer	SystemPort	The consumer port of the connection.
producer	SystemPort	The producer port of the connection.
priority	Number	Priority of the resulting orchestration rules (1 highest).

4.2 struct PdeSystem

A system of the plant.

Field	Type	Description
systemId	Identifier	Identifier of the system within the plant description.
systemName	Name	Name of the registered system (optional; metadata may identify it instead).
metadata	Metadata	Metadata the registered system must have.
ports	List<Port>	Services the system consumes or produces.

4.3 struct PlantDescriptionListResponse

List of plant descriptions.

Field	Type	Description
entries	List<PlantDescriptionResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.4 struct PlantDescriptionRequest

A plant description to create or replace.

Field	Type	Description
plantDescription	String	Name of the plant description.
active	Boolean	Make this the active plant description.
include	List<Number>	Identifiers of included plant descriptions.
systems	List<PdeSystem>	Systems to create or update.
connections	List<Connection>	Connections between consumer and producer ports.

4.5 struct PlantDescriptionResponse

A stored plant description.

Field	Type	Description
id	Number	Identifier of the plant description.
plantDescription	String	Name of the plant description.
active	Boolean	Whether this is the active plant description.
include	List<Number>	Identifiers of included plant descriptions.
systems	List<PdeSystem>	Systems to create or update.
connections	List<Connection>	Connections between consumer and producer ports.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.6 struct PlantDescriptionUpdate

Fields of a plant description to change; absent fields are kept.

Field	Type	Description
plantDescription	String	Name of the plant description.
active	Boolean	Make this the active plant description.
include	List<Number>	Identifiers of included plant descriptions.
systems	List<PdeSystem>	Systems to create or update.
connections	List<Connection>	Connections between consumer and producer ports.

4.7 struct Port

A service a plant system consumes or produces.

Field	Type	Description
portName	String	Name of the port within its system.
serviceDefinition	Name	Service definition name.
serviceInterface	String	Required interface template of the port, if any.
metadata	Metadata	Metadata the service instance must have.
consumer	Boolean	True for a consumed service, false for a produced one.

4.8 struct SystemPort

Reference to a port of a plant system.

Field	Type	Description
systemId	Identifier	Identifier of the system within the plant description.
portName	String	Name of the port within its system.

4.9 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “PlantDescriptionEngine – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-PlantDescriptionEngine-java-spring/documentation/v5.3.0/PlantDescriptionEngine-v5.3.0-SysD.pdf.

[2] ——, “plantDescriptionManagement HTTP/TLS/JSON – Interface Design Description (PlantDescriptionEngine),” Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-plantDescriptionManagement-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-23	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	